

VWO – Complete Test Case Specifications

Master Document | 187 Test Cases | Version 2.0 | July 2026

VWO – Test Case Master Index

Module	File	Total TC	P0	P1	P2	Automated
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VWO UI Total		103	48	43	12	84
Booking API – Health Check	api/TC-PNG-HealthCheck.md	6	4	2	0	6
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Booking API " Delete	api/TC-DEL-DeleteBooking.md	8	6	2	0	8
Booking API Total		61	32	29	0	61
Performance	nfr/TC-NFR-Performance.md	6	6	0	0	0
Security	nfr/TC-NFR-Security.md	6	4	2	0	0
Cross-Browser & Responsive	nfr/TC-NFR-CrossBrowser.md	6	3	3	0	0
Accessibility	nfr/TC-NFR-Accessibility.md	5	1	4	0	0
NFR Total		23	14	9	0	0
GRAND TOTAL		187	94	81	12	145

How to Use

- **P0**: Must pass before any release. Automated in CI pipeline.
- **P1**: Must pass before GA release. Automated in nightly regression.
- **P2**: Nice-to-have. Manual / exploratory.
- Each file uses the format: `TC-{MODULE}-{NNN}` for traceability to the VWO Test Plan.

Authentication & Access Control " Test Cases

File: TC-AUTH-Login.md | **Module:** Login | **Total:** 12 TC | **Automated:** 10

Ref	Priority	Automated
FR6, NFR	P0: 6, P1: 5, P2: 1	"... 10 / " 2 manual

TC-AUTH-001 " Valid Login with Email & Password

Field	Value
Priority	P0
FR ID	FR6
Automation	"... tests.login.LoginTest#testValidLogin
Precondition	User has valid credentials (ConfigReader.getUsername / getPassword)
Test Data	email = qa.automation@vwo.com , password = VWO@Test#2026
Steps	1. Navigate to https://app.vwo.com " 2. Enter email " 3. Enter password " 4. Click Login
Expected	Dashboard page loads with user avatar visible. URL contains /dashboard
Validation	dashboardPage.isDashboardDisplayed() == true

TC-AUTH-002 Invalid Password Shows Error

Field	Value
Priority	P0
FR ID	FR6
Automation	âœ… tests.login.LoginTest#testLoginWithInvalidPassword
Precondition	Known valid email, incorrect password
Test Data	email = qa.automation@vwo.com , password = WrongPassword@999
Steps	1. Enter email â†’ 2. Enter wrong password â†’ 3. Click Login
Expected	Error message displayed: "Invalid email or password" (or similar). User remains on login page.
Validation	loginPage.getErrorMessage() contains "Invalid" or "incorrect" or "password"

TC-AUTH-003 Unregistered Email Shows Error

Field	Value
Priority	P0
FR ID	FR6
Automation	âœ… tests.login.LoginTest#testLoginWithUnregisteredEmail
Precondition	Email does not exist in VWO system
Test Data	email = nonexistent@unknown.com , password = VWO@Test#2026
Steps	1. Enter unregistered email â†’ 2. Enter password â†’ 3. Click Login
Expected	Error: "Account not found" or "No account with this email"
Validation	Error message contains "not found" or "unrecognized"

TC-AUTH-004 Empty Credentials Blocked

Field	Value
Priority	P0
FR ID	FR6
Automation	âœ… tests.login.LoginTest#testLoginWithEmptyCredentials
Precondition	Login page loaded
Test Data	email = "" , password = ""
Steps	1. Leave email empty â†’ 2. Leave password empty â†’ 3. Click Login
Expected	Validation error: "Email is required" or equivalent. No network call made.

Validation	Error message contains "required" or "empty" or "email"
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TC-AUTH-005 **Password Field Masks Input**

Field	Value
Priority	P0
FR ID	NFR (Security)
Automation	âœ… tests.login.LoginTest#testPasswordFieldMasking
Precondition	Login page loaded
Steps	1. Type any password into password field
Expected	Characters appear as bullets/dots (â–•â–•â–•â–•â–•â–•â–•), not plain text
Validation	passwordField.getAttribute("type") equals "password"

TC-AUTH-006 **2FA Code Input Appears After Login**

Field	Value
Priority	P0
FR ID	NFR (Security)
Automation	âœ… Manual (requires 2FA-enabled test account)
Precondition	VWO account has 2FA enabled
Test Data	email/password for a 2FA-enabled account
Steps	1. Enter valid credentials â†’ 2. Click Login â†’ 3. Observe next screen
Expected	2FA code input field is displayed instead of Dashboard. User cannot bypass.
Validation	loginPage.is2FAFieldDisplayed() == true

TC-AUTH-007 **Invalid 2FA Code Blocks Access**

Field	Value
Priority	P1
FR ID	NFR (Security)
Automation	âœ… Manual
Precondition	2FA input field is visible
Test Data	2FA code = 000000 (wrong code)
Steps	1. Enter wrong 2FA code â†’ 2. Click Verify
Expected	Error: "Invalid verification code". User NOT redirected to dashboard.

Validation	Error toast visible, URL unchanged
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TC-AUTH-008 æœœ Forgot Password Sends Reset Email

Field	Value
Priority	P1
FR ID	FR6
Automation	âœœ... tests.login.LoginTest#testForgotPasswordFlow
Precondition	Login page loaded
Test Data	email = testuser@vwo.com
Steps	1. Click "Forgot Password?" â†’ 2. Enter registered email â†’ 3. Click "Send Reset Link"
Expected	Toast message: "Password reset link sent to your email"
Validation	Toast contains "sent" or "email" or "reset"

TC-AUTH-009 æœœ SSO Login Redirects to IdP

Field	Value
Priority	P1
FR ID	FR6
Automation	âœœ... tests.login.LoginTest
Precondition	Login page loaded, SSO configured for the test org
Steps	1. Click "Sign in with SSO" â†’ 2. Enter company domain
Expected	Redirected to identity provider (Okta/Azure AD). VWO login page no longer visible.
Validation	Current URL contains the IdP domain

TC-AUTH-010 æœœ Remember Me Persists Session

Field	Value
Priority	P0
FR ID	FR6
Automation	âœœ... tests.login.LoginTest#testRememberMeFunctionality
Precondition	Browser has no prior VWO session cookies
Test Data	Valid credentials
Steps	1. Check "Remember Me" â†’ 2. Login â†’ 3. Close browser â†’ 4. Re-open â†’ 5. Navigate to app.vwo.com

Expected	User is still logged in (session restored from cookie/token)
Validation	Dashboard loads without re-entering credentials

TC-AUTH-011 – Session Timeout Logs Out After Inactivity

Field	Value
Priority	P1
FR ID	NFR (Security)
Automation	☐ Manual (requires waiting 30 min)
Precondition	User is logged in
Steps	1. Login 2. Remain idle for 30 minutes 3. Click any navigation link
Expected	User is redirected to login page. Session token expired.
Validation	Login page is displayed, API returns 401

TC-AUTH-012 – Logout Clears Session

Field	Value
Priority	P0
FR ID	FR6
Automation	☑ tests.login.LoginTest#testLogoutFunctionality
Precondition	User is logged in (Dashboard visible)
Steps	1. Click user avatar 2. Click "Log out"
Expected	Returned to login page. Browser back button does NOT return to Dashboard.
Validation	loginPage.isLoginPageDisplayed() == true. Navigating back shows login page.

Dashboard – Test Cases

File: TC-DSH-Dashboard.md | Module: Dashboard | Total: 6 TC | Automated: 4

Ref	Priority	Automated
FR6, FR8	P0: 4, P1: 2	☑ 4 / ☐ 2

TC-DSH-001 – Dashboard Loads with Key Elements

Field	Value
Priority	P0

FR ID	FR6
Automation	âœ… tests.dashboard.DashboardTest#testDashboardKeyElements
Precondition	Valid login completed
Test Data	N/A
Steps	1. Login successfully â†’ 2. Observe dashboard page
Expected	Dashboard header visible. Active experiments count displayed. Recent activity panel visible. Sidebar navigation present. Notification bell visible.
Validation	dashboardPage.isDashboardDisplayed() == true AND dashboardPage.isRecentActivityVisible() == true

TC-DSH-002 âœ€ Recent Activity Panel Shows Latest Actions

Field	Value
Priority	P1
FR ID	FR6
Automation	âœ… Manual (requires prior experiment activity)
Precondition	At least one experiment was created/modified recently
Steps	1. Login â†’ 2. Observe "Recent Activity" panel on dashboard
Expected	Panel shows timestamped entries (e.g., "Experiment X launched 5 min ago"). Entries sorted newest-first.
Validation	Activity count > 0. Most recent entry timestamp <= current time.

TC-DSH-003 âœ€ Active Experiments Count Matches Actual

Field	Value
Priority	P0
FR ID	FR6
Automation	âœ… Manual (requires known experiment state)
Precondition	There are active (Running) experiments in the account
Steps	1. Login â†’ 2. Note the "Active Experiments" count â†’ 3. Navigate to Experiments list â†’ 4. Count Running experiments
Expected	Dashboard count equals the actual count of Running experiments.
Validation	Number(dashboard.activeCount) == experimentsPage.getRunningExperiments().size()

TC-DSH-004 âœ€ Sidebar Navigation to All Modules

Field	Value
Priority	P0
FR ID	FR6
Automation	âœ… tests.dashboard.DashboardTest#testNavigationToAllSections
Precondition	Dashboard loaded
Steps	1. Click each sidebar item: Experiments, Insights, Personalize, Reports, Plan, Integrations, Settings
Expected	Each click navigates to the correct page (page header matches). Active state is highlighted.
Validation	Each click â†’ correct URL pattern + page header text

TC-DSH-005 âœ€ Search Filters Experiments in Real-Time

Field	Value
Priority	P1
FR ID	FR6
Automation	âœ… tests.dashboard.DashboardTest#testDashboardSearch
Precondition	At least 2 experiments exist with distinct names
Test Data	Search term matching exactly one experiment name
Steps	1. Type partial experiment name in search bar â†’ 2. Observe experiment cards
Expected	Only experiments matching the search term are displayed. Cards filter as user types (debounced).
Validation	Visible card count â‰‰ total card count. All visible cards contain the search term.

TC-DSH-006 âœ€ Project Selector Filters Workspace Data

Field	Value
Priority	P1
FR ID	FR8
Automation	âœ… tests.dashboard.DashboardTest
Precondition	User has access to multiple projects
Test Data	Project name = "Default Project"
Steps	1. Click project selector dropdown â†’ 2. Select a different project
Expected	Dashboard reloads with data scoped to the selected project. Active experiments count updates.

Validation	Counts change after switching projects
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A/B Testing Test Cases

File: TC-ABT-ABTesting.md | Module: Experiments | Total: 18 TC | Automated: 16

Ref	Priority	Automated
FR1, FR2, FR3	P0: 10, P1: 6, P2: 2	16 / 2 manual

TC-ABT-001 Create A/B Test with Valid Data

Field	Value
Priority	P0
Precondition	Logged in, on Dashboard
Test Data	name="A/B Test - Homepage CTA Button", url="https://app.vwo.com/", hypothesis="Changing CTA button from blue to green will increase CTR by 15%"
Steps	1. Click "Create Experiment" â†’ "A/B Test" â†’ 2. Fill name, description, URL, hypothesis â†’ 3. Add variation "Green CTA Button" â†’ 4. Select primary metric "Click-Through Rate" â†’ 5. Add goal "Newsletter Sign-ups" â†’ 6. Save Draft
Expected	Experiment saved successfully as "Draft". Success toast shown. Experiment appears in experiments list.
Validation	experimentPage.isSuccessDisplayed() == true

TC-ABT-002 Create A/B Test Without Name Validation Error

Field	Value
Priority	P0
Steps	1. Create experiment with empty name â†’ 2. Click Save
Expected	Error: "Experiment name is required". Experiment NOT saved.
Validation	experimentPage.isErrorDisplayed() == true

TC-ABT-003 Add Multiple Variations (Control + Up to 10)

Field	Value
Priority	P0
Test Data	Control + 3 named variations
Steps	1. Create A/B test â†’ 2. Add 3 variations

Expected	Total 4 variations displayed (control + 3). Each variation block is editable.
Validation	<code>experimentPage.getVariationCount() == 4</code>

TC-ABT-004 ☞ Visual Editor (WYSIWYG) Variation Editing

Field	Value
Priority	P0
Steps	1. Create A/B test with variation â 2. Click "Visual Editor" tab â 3. Modify element (click + edit) â 4. Save
Expected	Visual editor loads the page in an iframe. Changes apply in real-time. Save persists changes.
Validation	<code>experimentPage.isSuccessDisplayed() == true</code>

TC-ABT-005 ☞ Code Editor Variation Editing (CSS/JS)

Field	Value
Priority	P1
Test Data	CSS: <code>body { background-color: #f5f5f5; }</code>
Steps	1. Create A/B test â 2. Add variation â 3. Click "Code Editor" â 4. Enter CSS/JS â 5. Apply
Expected	Code injected successfully. Preview reflects the CSS/JS changes.

TC-ABT-006 ☞ Toggle Between Visual and Code Editors

Field	Value
Priority	P1
Steps	1. Open Visual Editor â make change â 2. Switch to Code Editor â verify code reflects change â 3. Switch back
Expected	Changes sync between editors. No data loss on toggle.

TC-ABT-007 ☞ Set Primary Metric

Field	Value
Priority	P0
Test Data	Metric = "Click-Through Rate"
Steps	1. Create A/B test â 2. Select primary metric from dropdown
Expected	Metric selected. Goal configuration UI updates accordingly.

TC-ABT-008 ☞ Add Multiple Goals

Field	Value
Priority	P1
Steps	1. Create A/B test â€” 2. Add goal "Revenue per Visitor" â€” 3. Add goal "Time on Page"
Expected	Both goals listed. Primary metric distinct from secondary goals.

TC-ABT-009 â€” Launch Experiment â€” Status "Running"

Field	Value
Priority	P0
Steps	1. Create A/B test â€” Save Draft â€” 2. Click "Launch"
Expected	Status changes from "Draft" to "Running". Traffic starts flowing.
Validation	<code>experimentPage.getExperimentStatus().equalsIgnoreCase("running")</code>

TC-ABT-010 â€” Pause Running Experiment

Field	Value
Priority	P0
Steps	1. Launch experiment â€” 2. Click "Pause"
Expected	Status changes to "Paused". Visitors see control (original) only.
Validation	Status equals "paused"

TC-ABT-011 â€” Resume Paused Experiment

Field	Value
Priority	P0
Steps	1. Pause running experiment â€” 2. Click "Resume"
Expected	Status changes back to "Running". Existing data preserved, new visitors added.

TC-ABT-012 â€” Stop Experiment

Field	Value
Priority	P0
Steps	1. Launch experiment â€” 2. Click "Stop"
Expected	Status "Stopped". No more traffic allocated. Final results frozen.

TC-ABT-013 â€” Declare Winner After Statistical Significance

Field	Value
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Priority	P1
Precondition	Experiment has reached 95% confidence for one variation
Steps	1. Open experiment â†’ 2. Click "Declare Winner" â†’ 3. Select winning variation â†’ 4. Confirm
Expected	Selected variation marked as winner. All traffic now goes to winning variation.

TC-ABT-014 â€¢ Schedule Experiment with Dates

Field	Value
Priority	P1
Test Data	start="2026-01-15", end="2026-02-15"
Steps	1. Create A/B test â†’ 2. Click "Schedule" â†’ 3. Set start + end dates â†’ 4. Confirm
Expected	Schedule saved. Experiment auto-launches on start date, auto-stops on end date.

TC-ABT-015 â€¢ Preview Variations Across Devices

Field	Value
Priority	P1
Precondition	Experiment with at least one variation
Steps	1. Open experiment â†’ 2. Click "Preview" â†’ 3. Toggle between device sizes (Desktop/Mobile/Tablet)
Expected	Variation renders correctly on each device size. Device toggle changes viewport.

TC-ABT-016 â€¢ Set Traffic Allocation Percentage

Field	Value
Priority	P1
Test Data	Slider value = 50
Steps	1. In experiment config â†’ 2. Drag traffic allocation slider to 50% â†’ 3. Save
Expected	50% visitors see control, 50% see variation. Visual indicator updates.

TC-ABT-017 â€¢ Duplicate Existing Experiment

Field	Value
Priority	P2
Steps	1. Open existing experiment â†’ 2. Click "Duplicate" â†’ 3. Rename â†’ 4. Save
Expected	New experiment created with same config. Original unchanged.

TC-ABT-018 Delete Experiment in Draft Status

Field	Value
Priority	P2
Precondition	Experiment in "Draft" status
Steps	1. Open draft experiment â 2. Click "Delete" â 3. Confirm
Expected	Experiment removed from list. Cannot be recovered. Running experiments cannot be deleted.

Split URL Testing Test Cases

File: TC-SPL-SplitURL.md | Module: Split URL | Total: 4 TC | Automated: 3

TC-SPL-001 Create Split URL Test with 2+ Destination URLs

Field	Value
Priority	P0
Steps	1. Click "Create Experiment" â "Split URL" â 2. Enter control URL â 3. Add 2+ variation URLs â 4. Save
Expected	Experiment created with multiple URLs. Each variation maps to a different URL.
Validation	3 total variations (control + 2)

TC-SPL-002 Traffic Distribution Across Variations

Field	Value
Priority	P1
Steps	1. Create Split URL with 2 variations â 2. Set traffic allocation â 3. Launch
Expected	Traffic split evenly (or per configured %) across all URLs. Distribution UI shows percentages.

TC-SPL-003 Duplicate URL Validation

Field	Value
Priority	P1
Test Data	Variation URL = same as control URL
Steps	1. Create Split URL â 2. Add variation with identical URL to control
Expected	Warning/error: "URL must be different from control". Not allowed to save.

TC-SPL-004 Split URL with Different Domains

Field	Value
Priority	P1
Test Data	control=" https://example.com/page-a ", variation=" https://example.org/page-b "
Steps	1. Create Split URL with URLs on different domains
Expected	Test saves and launches. Cross-domain tracking works (cookies handled correctly).

Multivariate Testing – Test Cases

File: TC-MVT-Multivariate.md | Module: MVT | Total: 5 TC | Automated: 3

TC-MVT-001 – Create MVT with 4+ Combination Variations

Field	Value
Priority	P0
Test Data	2 headlines – 2 CTA colors = 4 combinations
Steps	1. Create MVT 2. Name experiment 3. Add 4 combination variations 4. Select primary metric 5. Save Draft
Expected	5 total entries (control + 4 combinations). All combinations listed.
Validation	experimentPage.getVariationCount() == 5

TC-MVT-002 – Preview All MVT Combinations

Field	Value
Priority	P1
Steps	1. Create MVT 2. Add combinations 3. Click "Preview"
Expected	Each combination renders correctly in preview. Switch between combos shows correct content.

TC-MVT-003 – Configure Multiple Goals for MVT

Field	Value
Priority	P1
Steps	1. Create MVT 2. Add primary metric + 2 secondary goals
Expected	Multiple goals configured. Results will show performance per goal per combination.

TC-MVT-004 – MVT Results Broken Down by Factor Interaction

Field	Value
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Priority	P2
Steps	1. Launch MVT â†’ 2. Wait for data â†’ 3. Open results
Expected	Results show main effects (per factor) AND interaction effects (combination). SmartStats calculates significance per factor.

TC-MVT-005 â€” MVT with > 8 Combinations

Field	Value
Priority	P2
Test Data	3 factors Ã— 3 levels = 9 combinations
Steps	1. Create MVT with 9+ combinations â†’ 2. Observe UI performance
Expected	UI handles large number of combinations without lag. All combinations previewable.

SmartStats Engine â€” Test Cases

File: TC-SS-SmartStats.md | Module: SmartStats | Total: 6 TC | Automated: 4

TC-SS-001 â€” SmartStats Panel Visible in Experiment Details

Field	Value
Priority	P0
Precondition	Experiment has collected visitor data
Steps	1. Open launched experiment â†’ 2. Scroll to results section
Expected	SmartStats panel visible. Shows Bayesian analysis with probability curves.

TC-SS-002 â€” Bayesian Confidence Score Displayed

Field	Value
Priority	P0
Steps	1. Open experiment with data â†’ 2. Observe confidence column
Expected	Each variation has a confidence score (percentage). Higher score = more likely to be better than control.
Validation	experimentPage.getConfidenceScore() is a non-empty percentage string

TC-SS-003 â€” Conversion Rate, Lift, and Visitors Count Shown

Field	Value
Priority	P0

Steps	1. Open experiment â†’ 2. View summary table
Expected	Columns: Variation, Visitors, Conversions, Conversion Rate, Lift, Confidence. All populated.

TC-SS-004 â€¢ Winner Automatically Suggested at 95% Confidence

Field	Value
Priority	P1
Precondition	Enough data collected for 95% confidence
Steps	1. Monitor experiment â†’ 2. When confidence reaches 95%
Expected	System highlights the winning variation. "Declare Winner" button becomes prominent. Suggested winner label shown.

TC-SS-005 â€¢ Manual Winner Override

Field	Value
Priority	P1
Steps	1. Open experiment â†’ 2. Click "Declare Winner" â†’ 3. Select a variation that is NOT the statistically suggested one
Expected	System allows manual override. Confirmation dialog: "Are you sure?" Winner declared as selected.

TC-SS-006 â€¢ SmartStats Data Refreshes in Real-Time

Field	Value
Priority	P1
Precondition	Experiment is running and receiving traffic
Steps	1. Open experiment results â†’ 2. Wait 30 seconds without refreshing
Expected	Visitor count and confidence score update automatically. No manual refresh needed.

Code Editor â€” Test Cases

File: TC-CDE-CodeEditor.md | Module: Code Editor | Total: 4 TC | Automated: 3

TC-CDE-001 â€¢ Code Editor Accepts CSS for Variation Changes

Field	Value
Priority	P0
Test Data	body { background-color: #f5f5f5; } .cta-button { color: #ffffff; background-color: #28a745; }
Steps	1. Open A/B test â†’ 2. Add variation â†’ 3. Click "Code Editor" â†’ 4. Enter CSS â†’ 5. Apply & Save

Expected	CSS injected. Preview shows the styled page. Changes persisted on save.
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TC-CDE-002 ¶ Code Editor Accepts JavaScript

Field	Value
Priority	P1
Test Data	document.querySelector('.hero-title').innerText = 'Welcome to Optimized VWO';
Steps	1. Open Code Editor ¶ 2. Enter JS ¶ 3. Apply
Expected	JavaScript executes in the preview. DOM changes visible.

TC-CDE-003 ¶ Toggle Between Visual and Code Editors

Field	Value
Priority	P1
Steps	1. Open Visual Editor ¶ make a change ¶ 2. Switch to Code Editor ¶ 3. Switch back to Visual Editor
Expected	Code changes visible when switching to Code Editor. Visual Editor reflects the code. No data corruption on toggle.

TC-CDE-004 ¶ Syntax Error Handling in Code Editor

Field	Value
Priority	P1
Test Data	Malformed CSS: body { background-color: ; }
Steps	1. Open Code Editor ¶ 2. Enter invalid CSS/JS ¶ 3. Apply
Expected	Error message: "Invalid syntax at line X". Code NOT applied. Variation retains last valid state.

Audience Targeting ¶ Test Cases

File: TC-AUD-Audience.md | Module: Audience | Total: 6 TC | Automated: 5

TC-AUD-001 ¶ Target by Geography

Field	Value
Priority	P0
Test Data	Geography = "United States"

Steps	1. Create A/B test â†’ 2. Go to Audience section â†’ 3. Add condition: Geography = United States â†’ 4. Save
Expected	Audience condition saved. Only US visitors will see the experiment.

TC-AUD-002 â€¢ Target by Device Type

Field	Value
Priority	P0
Test Data	Device = "Mobile"
Steps	1. Add audience condition: Device = Mobile â†’ 2. Save
Expected	Only mobile visitors see the experiment.

TC-AUD-003 â€¢ Target by Behavior (Returning vs New)

Field	Value
Priority	P0
Test Data	Behavior = "Returning Visitor"
Steps	1. Add audience condition: Behavior = Returning Visitor â†’ 2. Save
Expected	Only logged-in returning visitors see the experiment. New visitors see control.

TC-AUD-004 â€¢ Target by Custom Attribute

Field	Value
Priority	P1
Test Data	Custom attribute: plan_type = enterprise
Steps	1. Add audience condition â†’ Custom Attribute â†’ 2. Set key=plan_type, value=enterprise â†’ 3. Save
Expected	Condition saved. Only visitors with matching custom attribute are targeted.

TC-AUD-005 â€¢ Combine Multiple Conditions (AND Logic)

Field	Value
Priority	P1
Test Data	Geography=US AND Device=Desktop AND Behavior=New Visitor
Steps	1. Add 3 audience conditions â†’ 2. Save

Expected	All conditions apply with AND logic. Audience preview shows reduced reach. Narrows down to visitors matching ALL conditions.
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TC-AUD-006 – Audience Preview Shows Estimated Reach

Field	Value
Priority	P1
Steps	1. Add audience condition 2. Observe audience preview panel
Expected	Preview shows: "Estimated reach: X% of all visitors (Y visitors/month)". Reach updates in real-time as conditions change.

Heatmaps – Test Cases

File: TC-HTM-Heatmaps.md | **Module:** Insights - Heatmaps | **Total:** 6 TC | **Automated:** 5

TC-HTM-001 – Generate Click Heatmap for Valid Page URL

Field	Value
Priority	P0
Test Data	URL = " https://app.vwo.com/ ", Type = "Click Heatmap"
Steps	1. Navigate to Insights 2. Enter page URL 3. Select "Click Heatmap" 4. Click "Generate"
Expected	Heatmap overlay renders on the page. Click hotspots color-coded (red=most clicks, blue=least). Tooltip shows click count on hover.
Validation	insightsPage.isHeatmapDisplayed() == true

TC-HTM-002 – Generate Scroll Heatmap with Depth Overlay

Field	Value
Priority	P0
Steps	1. Select "Scroll Heatmap" 2. Enter URL 3. Generate
Expected	Scroll depth overlay shows % of visitors who scrolled to each page section. "Fold" line visible at common breakpoints.

TC-HTM-003 – Generate Move Heatmap

Field	Value
Priority	P1
Steps	1. Select "Move Heatmap" 2. Generate

Expected	Mouse movement trails visualized. Areas with most cursorâ€œ™ highlighted.
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TC-HTM-004 â€œ Filter Heatmap by Device Type

Field	Value
Priority	P1
Steps	1. Generate heatmap â€ 2. Filter by "Mobile" â€ 3. Filter by "Desktop"
Expected	Heatmap refreshes per device. Mobile heatmap shows different click patterns (thumb zone). Desktop shows wider spread.

TC-HTM-005 â€œ Invalid URL Shows Graceful Error

Field	Value
Priority	P1
Test Data	URL = " https://invalid-page-12345.com "
Steps	1. Enter unreachable URL â€ 2. Generate
Expected	Error: "Unable to load page. Please verify the URL is accessible." Heatmap NOT generated.
Validation	insightsPage.isHeatmapDisplayed() == false

TC-HTM-006 â€œ Heatmap Overlay Disappears on Exiting Heatmap View

Field	Value
Priority	P2
Steps	1. Generate heatmap â€ 2. Close heatmap view / navigate away
Expected	Page returns to normal. No residual overlay elements. Original page layout unchanged.

Session Recordings â€” Test Cases

File: TC-REC-SessionRecordings.md | Module: Insights - Recordings | Total: 5 TC | Automated: 4

TC-REC-001 â€œ Session Recordings List Loads with Metadata

Field	Value
Priority	P0
Steps	1. Navigate to Insights â€ Session Recordings
Expected	List of recorded sessions displayed. Each entry shows: User ID, Date, Duration, Pages visited, Actions count. Pagination if > 20.
Validation	insightsPage.getSessionRecordingCount() >= 0

TC-REC-002 – Play Session Recording Opens Player

Field	Value
Priority	P0
Steps	1. Click a recording 2. Click "Play"
Expected	Recording player opens. Screen replay starts automatically. Timeline progress bar visible.

TC-REC-003 – Playback Controls Functional

Field	Value
Priority	P1
Steps	1. Play recording 2. Click Pause 3. Click Play 4. Change speed (2x, 4x) 5. Skip forward/backward
Expected	Pause: replay freezes. Play: resumes. Speed: replay accelerates audibly. Skip: jumps to that timeline position.

TC-REC-004 – Filter Recordings by Date Range

Field	Value
Priority	P1
Test Data	Date range = "2026-01-01 to 2026-01-31"
Steps	1. Click date filter 2. Select start/end dates 3. Apply
Expected	List filtered to recordings within the date range. Count updates. Empty state if no recordings.

TC-REC-005 – Filter Recordings by User ID

Field	Value
Priority	P1
Test Data	Known user ID from a previous test session
Steps	1. Enter user ID in search 2. Apply
Expected	Only recordings from that user displayed. Exact match required.

Funnels & Surveys – Test Cases

File: TC-FUN-FunnelsSurveys.md | Module: Insights - Funnels & Surveys | Total: 5 TC | Automated: 5

TC-FUN-001 – Create Funnel with 2+ Steps

Field	Value
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Priority	P0
Test Data	Step URLs: https://app.vwo.com/ â https://app.vwo.com/pricing â https://app.vwo.com/signup
Steps	1. Navigate to Insights â Funnels â 2. Click "Create Funnel" â 3. Add step URLs in order â 4. Save
Expected	Funnel created with all steps. Steps ordered correctly. Save confirmation shown.
Validation	insightsPage.getFunnelStepCount() >= 2

TC-FUN-002 â Funnel Shows Drop-Off % at Each Step

Field	Value
Priority	P0
Precondition	Funnel has collected visitor data
Steps	1. Open funnel â 2. Observe visualization
Expected	Each step shows: Visitors entering, % drop-off from previous step, % continued. Visual bar/arrow indicating flow. Highest drop-off clearly highlighted.

TC-FUN-003 â Create and Publish On-Page Survey

Field	Value
Priority	P1
Test Data	Question = "How likely are you to recommend VWO to a colleague?"
Steps	1. Navigate to Insights â Surveys â 2. Click "Create Survey" â 3. Enter question â 4. Configure targeting â 5. Publish
Expected	Survey created and published. Survey appears on target pages. Count increases by 1.

TC-FUN-004 â Survey Responses Collected in Dashboard

Field	Value
Priority	P2
Precondition	Survey is published and receiving responses
Steps	1. Open published survey â 2. View "Responses" tab
Expected	Response count > 0. Individual responses listed with timestamps. Aggregate data (NPS score, distribution chart).

TC-FUN-005 â Export Funnel Data to CSV

Field	Value
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Priority	P1
Steps	1. Open funnel â€ 2. Click "Export" â€ 3. Select CSV format
Expected	CSV file downloads. File contains: Step name, Visitors, Drop-off count, Drop-off %, Conversion %. Data matches UI.

Personalization â€” Test Cases

File: TC-PER-Personalization.md | Module: Personalize | Total: 10 TC | Automated: 8

TC-PER-001 â€¢ Create Personalization Campaign with Segments

Field	Value
Priority	P0
Test Data	Name="US Visitor Personalized Banner", Segment=Geography=US, Content="Welcome to VWO - Exclusive US Offer!"
Steps	1. Navigate to Personalize â€ 2. Click "Create Campaign" â€ 3. Enter name â€ 4. Add segment â€ 5. Add personalized content â€ 6. Save Draft
Expected	Campaign saved as "Draft". Appears in campaign list.

TC-PER-002 â€¢ Geographic Segment Targeting

Field	Value
Priority	P0
Test Data	Segment = Geography: "United States"
Steps	1. Create campaign â€ 2. Add geo segment
Expected	Segment configured. Audience preview shows US-only reach.

TC-PER-003 â€¢ Behavioral Segment Targeting

Field	Value
Priority	P0
Test Data	Segment = Behavior: "Returning Visitor"
Steps	1. Create campaign â€ 2. Add behavioral segment â€ 3. Add content â€ 4. Launch
Expected	Campaign launched. Only returning visitors see personalized content.

TC-PER-004 â€¢ Device-Based Segment Targeting

Field	Value
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Priority	P1
Test Data	Segment = Device: "Mobile"
Steps	1. Create campaign â†’ 2. Add device segment â†’ 3. Save
Expected	Campaign targets mobile users only.

TC-PER-005 â€¢ Configure Personalized Content

Field	Value
Priority	P0
Test Data	Content = "Welcome to VWO - Exclusive US Offer!"
Steps	1. Create campaign â†’ 2. Enter personalized content in editor (text/HTML/CSS)
Expected	Content saved. Content renders correctly in preview.

TC-PER-006 â€¢ Preview Personalization Before Launch

Field	Value
Priority	P1
Steps	1. Create campaign with content â†’ 2. Click "Preview"
Expected	Preview panel shows the page with personalized content applied. "View as different segments" toggle works.
Validation	<code>personalizePage.isPersonalizedPreviewDisplayed() == true</code>

TC-PER-007 â€¢ Launch Campaign â†’ Status "Running"

Field	Value
Priority	P0
Precondition	Campaign has segments + content configured
Steps	1. Click "Launch"
Expected	Status changes from "Draft" to "Running". Campaign starts serving personalized content.

TC-PER-008 â€¢ Pause Active Campaign

Field	Value
Priority	P1
Steps	1. Open running campaign â†’ 2. Click "Pause"
Expected	Status changes to "Paused". Original (non-personalized) content served to all visitors.

TC-PER-009 Set Traffic Percentage for Campaign

Field	Value
Priority	P1
Test Data	Traffic % = 50
Steps	1. Create campaign 2. Set traffic percentage slider to 50% 3. Save
Expected	50% of qualifying visitors see personalized content. 50% see original.

TC-PER-010 Campaign with Multiple Segments

Field	Value
Priority	P1
Test Data	Geo=US + Behavior=New Visitor + Device=Desktop
Steps	1. Create campaign 2. Add 3 segments (geo + behavior + device) 3. Save
Expected	All 3 segments combined with AND logic. Reach narrowed to intersection. Campaign saved successfully.

Reports & Dashboards Test Cases

File: TC-RPT-Reports.md | Module: Reports | Total: 10 TC | Automated: 10

TC-RPT-001 Reports Page Loads with Summary Statistics

Field	Value
Priority	P0
Steps	1. Navigate to Reports
Expected	Reports dashboard loads. Summary cards visible: Active Experiments, Total Visitors, Avg Conversion Rate, Revenue Impact.
Validation	reportsPage.isReportsPageDisplayed() == true

TC-RPT-002 Experiment Performance Table

Field	Value
Priority	P0
Steps	1. Open Reports 2. View experiment table
Expected	Table columns: Name, Variations, Visitors, Conversions, Conversion Rate, Confidence, Lift, Revenue. Sortable by each column. Paginated.

Validation	Headers contain "Name", "Conversion", "Visitors"
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TC-RPT-003 Æ Filter Reports by Status

Field	Value
Priority	P1
Test Data	Filter = "Running"
Steps	1. Click status filter Æ 2. Select "Running" Æ 3. Apply
Expected	Table shows only Running experiments. Count updates. Other statuses hidden.

TC-RPT-004 Æ Filter Reports by Project

Field	Value
Priority	P1
Test Data	Project = "Default Project"
Steps	1. Click project filter Æ 2. Select project Æ 3. Apply
Expected	Data scoped to selected project. All metrics update.

TC-RPT-005 Æ Filter Reports by Date Range

Field	Value
Priority	P1
Test Data	Range = "2026-01-01 to 2026-01-31"
Steps	1. Open date picker Æ 2. Select start + end Æ 3. Apply
Expected	Charts + tables refresh to show data only within the date range.

TC-RPT-006 Æ Chart Visualizations Render

Field	Value
Priority	P1
Steps	1. Open Reports Æ 2. Observe chart area
Expected	Charts render: Conversion trend line, Experiment performance bar chart, Traffic distribution pie chart. Interactive (hover shows tooltip).
Validation	reportsPage.isChartDisplayed() == true

TC-RPT-007 Æ Export Report to CSV

Field	Value
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Priority	P1
Steps	1. Click "Export" â†’ 2. Select "CSV"
Expected	File downloads with .csv extension. Contains all current table data (respecting filters). Columns match UI.

TC-RPT-008 â€¢ Export Report to PDF

Field	Value
Priority	P1
Steps	1. Click "Export" â†’ 2. Select "PDF"
Expected	PDF downloads. Contains Summary cards + Charts + Table. Formatted for printing.

TC-RPT-009 â€¢ Schedule Recurring Report Delivery

Field	Value
Priority	P2
Test Data	Recipients = " ga-team@vwo.com ", Frequency = "Weekly"
Steps	1. Click "Schedule Report" â†’ 2. Enter email recipients â†’ 3. Select frequency (Daily/Weekly/Monthly) â†’ 4. Confirm
Expected	Scheduled report created. Confirmation: "Report scheduled successfully." Email sent at configured frequency.

TC-RPT-010 â€¢ Conversion Summary Shows Lift/Decline Trends

Field	Value
Priority	P1
Steps	1. Open Reports â†’ 2. View conversion summary card
Expected	Summary shows: Current conversion rate, Change vs previous period (â†’/â†“ arrow + percentage), Trend indicator (positive/negative/neutral).

Integrations â€” Test Cases

File: TC-INT-Integrations.md | **Module:** Integrations | **Total:** 3 TC | **Automated:** 2

TC-INT-001 â€¢ Navigate to Integrations Listing Page

Field	Value
Priority	P1

Steps	1. Click "Integrations" in sidebar
Expected	Integrations page loads. Available connectors listed with search/filter. Categories: Analytics, CRM, Commerce, Data Platforms.

TC-INT-002 – Search for Integration Connector

Field	Value
Priority	P2
Test Data	Search term = "Salesforce"
Steps	1. On Integrations page 2. Type connector name in search
Expected	Matching connectors filtered in real-time. Partial match works ("Sales" finds "Salesforce"). No results message if not found.

TC-INT-003 – Connect to Google Analytics

Field	Value
Priority	P2
Steps	1. Click Google Analytics connector 2. Click "Connect" 3. Authorize via OAuth 4. Select GA property 5. Save
Expected	Connection established. Status shows "Connected". Data sync begins. Experiments created in VWO appear in GA as annotations.

Plan Management – Test Cases

File: TC-PLN-PlanManagement.md | Module: Plan | Total: 3 TC | Automated: 2

TC-PLN-001 – Access Plan/Program Management Dashboard

Field	Value
Priority	P1
Steps	1. Click "Plan" in sidebar
Expected	Plan dashboard loads. Kanban-style board visible. Columns: Backlog, To Do, In Progress, Review, Done. Experiment cards in each column.

TC-PLN-002 – Create Experiment Backlog Task

Field	Value
Priority	P2

Test Data	Task name = "Test homepage hero section redesign"
Steps	1. Click "Add Task" in Backlog column â†’ 2. Enter title â†’ 3. Assign owner â†’ 4. Set priority â†’ 5. Add description/hypothesis â†’ 6. Save
Expected	Task card appears in Backlog column. Card shows title, owner avatar, priority badge.

TC-PLN-003 â€¢ Kanban Column Drag and Drop

Field	Value
Priority	P2
Steps	1. Drag a task card from "Backlog" â†’ 2. Drop into "To Do" column
Expected	Card moves to "To Do". Status updated. Column counts update. Change persists on page refresh.

Booking API â€™ Health Check â€™ Test Cases

File: api/TC-PNG-HealthCheck.md | **Endpoint:** GET /ping | **Total:** 6 TC | **Automated:** 6

TC-PNG-001 â€¢ Health Check Returns 201 Created

Field	Value
Priority	P0
Steps	GET https://restful-booker.herokuapp.com/ping
Expected	Status: 201 Created. Body: empty.

TC-PNG-002 â€¢ Response Time Under 500ms

Field	Value
Priority	P0
Steps	GET /ping â€™ measure response time
Expected	response.time < 500ms

TC-PNG-003 â€¢ Responds Without Authentication

Field	Value
Priority	P0
Steps	GET /ping (no Auth header)
Expected	Status: 201. Endpoint is public.

TC-PNG-004 â€¢ Rejects POST/PUT/DELETE (405)

Field	Value
Priority	P1
Steps	POST /ping, PUT /ping, DELETE /ping, PATCH /ping
Expected	Each returns 405 Method Not Allowed.

TC-PNG-005 â€¢ Consecutive 10 Rapid Calls

Field	Value
Priority	P1
Steps	Call GET /ping 10 times in rapid succession
Expected	All 10 return 201. No rate limiting. No failures.

TC-PNG-006 â€¢ Response Headers Present

Field	Value
Priority	P1
Steps	GET /ping â€” inspect headers
Expected	Headers present: Content-Type, Server, Date, Content-Length

Booking API â€” Authentication â€” Test Cases

File: api/TC-AUTH-ApiAuth.md | Endpoint: POST /auth | Total: 5 TC | Automated: 5

TC-AUTH-020 â€¢ Valid Credentials Return 200 with Token

Field	Value
Priority	P0
Test Data	{"username": "admin", "password": "password123"}
Steps	POST /auth with valid JSON body
Expected	Status 200. Response: {"token": "<non-empty-string>"}

TC-AUTH-021 â€¢ Token is Non-Empty String

Field	Value
Priority	P0
Steps	Extract token from auth response

Expected	token is not null, not empty, and is a string.
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TC-AUTH-022 Invalid Credentials Handling

Field	Value
Priority	P1
Test Data	{"username": "invalid", "password": "invalid"}
Steps	POST /auth with bad credentials
Expected	Status 200 but token may not work for write operations. System behavior documented.

TC-AUTH-023 Empty Credentials

Field	Value
Priority	P1
Test Data	{"username": "", "password": ""}
Steps	POST /auth with empty fields
Expected	Status 400 Bad Request or 200. Error message if 400.

TC-AUTH-024 Missing Content-Type Header

Field	Value
Priority	P1
Steps	Send POST /auth with JSON body but NO Content-Type: application/json header
Expected	Status 400 Bad Request OR 415 Unsupported Media Type.

Booking API “ Get All Bookings ” Test Cases

File: api/TC-BOK-GetAllBookings.md | Endpoint: GET /booking | Total: 10 TC | Automated: 10

TC-BOK-001 Returns 200 OK with Booking IDs Array

Field	Value
Priority	P0
Steps	GET /booking
Expected	Status 200. Body: [{"bookingid": 1}, {"bookingid": 2}, ...]

TC-BOK-002 Non-Empty JSON Array

Field	Value
Priority	P0
Steps	GET /booking
Expected	Response is a JSON array with at least 1 element.

TC-BOK-003 â€¢ bookingid is Positive Integer

Field	Value
Priority	P0
Steps	GET /booking â€” validate each bookingid
Expected	Every bookingid is a positive integer > 0.

TC-BOK-004 â€¢ Booking IDs Are Unique

Field	Value
Priority	P1
Steps	Extract all bookingid values
Expected	Count of distinct IDs == total count. No duplicates.

TC-BOK-005 â€¢ Filter by firstname

Field	Value
Priority	P1
Test Data	?firstname=John
Steps	GET /booking?firstname=John
Expected	Status 200. Results contain only bookings where firstname == "John".

TC-BOK-006 â€¢ Filter by lastname

Field	Value
Priority	P1
Test Data	?lastname=Smith

TC-BOK-007 â€¢ Filter by checkin Date

Field	Value
Priority	P1
Test Data	?checkin=2026-01-01

Expected	Status 200. Results have bookingdates.checkin >= 2026-01-01.
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TC-BOK-008 ðŸ“œ Filter by checkout Date

Field	Value
Priority	P1

TC-BOK-009 ðŸ“œ Combined Filters

Field	Value
Priority	P1
Expected	Results match BOTH conditions (AND logic).

TC-BOK-010 ðŸ“œ Invalid Filter Returns Empty Array

Field	Value
Priority	P1
Test Data	?firstname=NonExistentName_XYZ
Expected	Status 200. Body: [] (empty array).

Booking API ðŸ“œ Get Single Booking ðŸ“œ Test Cases

File: api/TC-BOK-GetSingleBooking.md | Endpoint: GET /booking/{id} | Total: 12 TC | Automated: 12

TC-BOK-011 ðŸ“œ Valid ID Returns 200 with Full Booking

Field	Value
Priority	P0
Steps	GET /booking/1
Expected	Status 200. Full booking object returned.

TC-BOK-012 ðŸ“œ All Required Fields Present

Field	Value
Priority	P0
Expected	Fields: firstname, lastname, totalprice, depositpaid, bookingdates (with checkin, checkout), additionalneeds. None null.

TC-BOK-013 ðŸ“œ Name Fields Non-Empty

Field	Value
Priority	P0
Expected	firstname.trim().length() > 0, lastname.trim().length() > 0

TC-BOK-014 â€¢ totalprice is Positive

Field	Value
Priority	P0
Expected	totalprice > 0

TC-BOK-015 â€¢ depositpaid is Boolean

Field	Value
Priority	P0
Expected	`depositpaid == true

TC-BOK-016 â€¢ Booking Dates in YYYY-MM-DD Format

Field	Value
Priority	P0
Expected	checkin and checkout match regex: \d{4}-\d{2}-\d{2}

TC-BOK-017 â€¢ checkout >= checkin

Field	Value
Priority	P0
Expected	checkout >= checkin (string comparison on YYYY-MM-DD is valid)

TC-BOK-018 â€¢ Non-Existent ID Returns 404

Field	Value
Priority	P0
Steps	GET /booking/99999
Expected	Status 404 Not Found

TC-BOK-019 â€¢ Negative ID Returns 400/404

Field	Value
Priority	P1

Steps	GET /booking/-1
Expected	Status 400 or 404

TC-BOK-020 â€¢ String ID Returns 400/404

Field	Value
Priority	P1
Steps	GET /booking/abc
Expected	Status 400 or 404

TC-BOK-021 â€¢ Special Character ID Returns 400/404

Field	Value
Priority	P1
Steps	GET /booking/@#\$
Expected	Status 400 or 404

TC-BOK-022 â€¢ Response Defaults to JSON

Field	Value
Priority	P1
Steps	GET /booking/1 (no Accept header)
Expected	Status 200. Content-Type contains application/json.

Booking API â€” Full Update (PUT) â€” Test Cases

File: api/TC-PUT-UpdateBooking.md | Endpoint: PUT /booking/{id} | Total: 10 TC | Automated: 10

TC-PUT-001 â€¢ Full Update with Valid Auth Returns 200

Field	Value
Priority	P0
Test Data	Auth = Basic admin:password123, Body = { "firstname": "Jane", "lastname": "Doe", "totalprice": 500, "depositpaid": false, "bookingdates": { "checkin": "2026-08-01", "checkout": "2026-08-10", "additionalneeds": "Lunch" } }
Steps	1. Create booking â†’ 2. PUT /booking/{id} with full payload + auth
Expected	Status 200. Updated values returned in response.

TC-PUT-002 – All Fields Updated Correctly

Field	Value
Priority	P0
Steps	PUT â GET booking â compare
Expected	Every field matches the PUT payload. Round-trip verified.

TC-PUT-003 – Response Body Matches Request (Round-Trip)

Field	Value
Priority	P0
Expected	PUT response body == request body (all fields).

TC-PUT-004 – Update Without Auth Returns 403

Field	Value
Priority	P0
Steps	PUT /booking/{id} with valid body but NO Auth header
Expected	Status 403 Forbidden.

TC-PUT-005 – Update with Invalid Auth Returns 403

Field	Value
Priority	P1
Steps	PUT /booking/{id} with Base64("bad:creds")
Expected	Status 403 or 401.

TC-PUT-006 – Update to Non-Existent ID

Field	Value
Priority	P1
Steps	PUT /booking/99999 with valid auth + body
Expected	Status 404 or 405.

TC-PUT-007 – Missing Required Fields

Field	Value
Priority	P1
Test Data	{"firstname": "OnlyFirst"} (missing lastname, totalprice, etc.)

Expected	Status 400 Bad Request OR 200 (depending on API design â€” document behavior).
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TC-PUT-008 â€” Invalid Data Types

Field	Value
Priority	P1
Test Data	"totalprice": "not-a-number"
Expected	Status 400 or 500. Type mismatch should be rejected.

TC-PUT-009 â€” Extra Unknown Fields Accepted (Tolerant)

Field	Value
Priority	P1
Test Data	Extra fields: "extraField": "shouldBeIgnored"
Expected	Status 200. Extra fields ignored (not persisted). Backward compatible.

TC-PUT-010 â€” checkout Before checkin

Field	Value
Priority	P1
Test Data	checkin="2026-02-01", checkout="2026-01-01"
Expected	Status 400 OR system auto-corrects.

Booking API â€” Partial Update (PATCH) â€” Test Cases

File: api/TC-PTC-PartialUpdate.md | **Endpoint:** PATCH /booking/{id} | **Total:** 10 TC | **Automated:** 10

TC-PTC-001 â€” Patch Single Field Returns 200

Field	Value
Priority	P0
Test Data	{"firstname": "UpdatedName"}
Steps	Create booking â†’ PATCH /booking/{id} with single field + auth
Expected	Status 200. firstname updated.

TC-PTC-002 â€” Only Patched Field Changes, Others Unchanged

Field	Value
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Priority	P0
Steps	GET before â€˜ PATCH firstname â€˜ GET after
Expected	firstname changed. lastname, totalprice, depositpaid, bookingdates, additionalneeds all identical to pre-patch values.

TC-PTC-003 â€¢ Patch totalprice

Field	Value
Priority	P0
Test Data	{"totalprice": 999}
Expected	totalprice updated to 999.

TC-PTC-004 â€¢ Patch bookingdates.checkin

Field	Value
Priority	P0
Test Data	{"bookingdates": {"checkin": "2026-12-01"}}
Expected	Only checkin changes. checkout stays same.

TC-PTC-005 â€¢ Patch Without Auth Returns 403

Field	Value
Priority	P0
Expected	Status 403 Forbidden.

TC-PTC-006 â€¢ Empty Body Returns 400

Field	Value
Priority	P1
Test Data	{}
Expected	Status 400 or 200 (idempotent).

TC-PTC-007 â€¢ Patch Non-Existent ID Returns 404

Field	Value
Priority	P1
Steps	PATCH /booking/99999
Expected	Status 404 or 405.

TC-PTC-008 Invalid Field Name Ignored

Field	Value
Priority	P1
Test Data	{"nonexistentField": "shouldBeIgnored"}
Expected	Status 200. No error. Field ignored.

TC-PTC-009 Null Value for Required Field

Field	Value
Priority	P1
Test Data	{"firstname": null}
Expected	Status 400 (reject null for required fields).

TC-PTC-010 additionalneeds Set to Empty String

Field	Value
Priority	P1
Test Data	{"additionalneeds": ""}
Expected	Status 200. additionalneeds is empty string.

Booking API Delete Booking Test Cases

File: api/TC-DEL-DeleteBooking.md | Endpoint: DELETE /booking/{id} | Total: 8 TC | Automated: 8

TC-DEL-001 Delete with Valid Auth Returns 201

Field	Value
Priority	P0
Steps	Create booking â DELETE /booking/{id} with auth
Expected	Status 201 Created.

TC-DEL-002 Deleted Booking Returns 404 on GET

Field	Value
Priority	P0
Steps	DELETE â GET same ID
Expected	DELETE: 201. GET: 404 Not Found.

TC-DEL-003 Delete Without Auth Returns 403

Field	Value
Priority	P0
Steps	DELETE /booking/{id} (no Auth header)
Expected	Status 403 Forbidden.

TC-DEL-004 Delete with Invalid Auth Returns 403

Field	Value
Priority	P0
Steps	DELETE /booking/{id} with Base64("bad:creds")
Expected	Status 403 or 401.

TC-DEL-005 Delete Non-Existent ID Returns 405

Field	Value
Priority	P1
Steps	DELETE /booking/99999 with auth
Expected	Status 405 Method Not Allowed.

TC-DEL-006 Double Delete Returns 405

Field	Value
Priority	P1
Steps	DELETE same ID twice
Expected	First: 201. Second: 405 or 404.

TC-DEL-007 GET on Delete Endpoint Returns 200

Field	Value
Priority	P0
Steps	GET /booking/{id} (verifying method safety)
Expected	Status 200.

TC-DEL-008 Create Delete Verify Cycle

Field	Value
Priority	P1

Steps	POST (create) â†’ GET (verify exists) â†’ DELETE â†’ GET (verify gone)
Expected	Full lifecycle works. No orphaned data.

Non-Functional â€” Performance â€” Test Cases

File: nfr/TC-NFR-Performance.md | Module: Performance | Total: 6 TC | Automated: 0

TC-PRF-01 â€¢ Dashboard Page Load Time < 2s

Field	Value
Priority	P0
Precondition	User logged in, cache cleared
Steps	1. Start timer â†’ 2. Navigate to Dashboard â†’ 3. Wait for page fully loaded (all XHRs complete) â†’ 4. Stop timer
Expected	Total load time < 2000ms (2 seconds).
Tool	Selenium + Stopwatch or Navigation Timing API

TC-PRF-02 â€¢ Experiment Creation/Save < 2s

Field	Value
Priority	P0
Steps	1. Create experiment â†’ 2. Measure time from "Save" click to success toast
Expected	< 2000ms for draft save.

TC-PRF-03 â€¢ Heatmap Generation < 5s

Field	Value
Priority	P0
Steps	1. Enter URL â†’ 2. Click "Generate Heatmap" â†’ 3. Measure until overlay renders
Expected	< 5000ms (5 seconds).

TC-PRF-04 â€¢ Report Page Load with 100+ Experiments < 3s

Field	Value
Priority	P0
Precondition	Workspace has 100+ experiments
Steps	1. Navigate to Reports â†’ 2. Measure full page load time

Expected	< 3000ms (3 seconds). Pagination/no-data states handle gracefully.
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TC-PRF-05 API Response Time < 500ms

Field	Value
Priority	P0
Steps	Call each API endpoint and measure response time
Criteria	GET /booking < 500ms, GET /booking/{id} < 500ms, POST /booking < 1000ms

TC-PRF-06 Concurrent Experiment Launches (50 Simultaneous)

Field	Value
Priority	P0
Steps	1. Create 50 draft experiments 2. Launch all simultaneously via API
Expected	No 500 errors. All 50 launch successfully. System recovers within 30s.
Tool	JMeter / Locust

Non-Functional Security Test Cases

File: nfr/TC-NFR-Security.md | Module: Security | Total: 6 TC | Automated: 0

TC-SEC-01 2FA Enforcement for Enterprise Accounts

Field	Value
Priority	P0
Precondition	VWO account with 2FA enabled
Steps	1. Login with valid credentials 2. Observe post-login screen
Expected	2FA code input MANDATORY. User cannot skip/ bypass. Dashboard NOT accessible without 2FA.

TC-SEC-02 RBAC Admin vs Viewer Permissions

Field	Value
Priority	P0
Test Data	Two accounts: Admin role + Viewer role
Steps	1. Login as Admin can create/launch/delete experiments 2. Login as Viewer observe options

Expected	Admin: full CRUD. Viewer: read-only (no Create/Edit/Delete buttons). API returns 403 for write operations.
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TC-SEC-03 Session Timeout After 30 Minutes

Field	Value
Priority	P0
Steps	1. Login 2. Wait 30 min without interaction 3. Click any navigation link
Expected	Redirected to login page. Session token invalidated. API returns 401.

TC-SEC-04 XSS Injection Prevention

Field	Value
Priority	P1
Test Data	Payload: <script>alert('XSS')</script> in firstname, additionalneeds
Steps	1. Create booking with XSS payload in string fields 2. GET booking 3. Render in UI
Expected	API stores payload as-is. UI renders as escaped text (NOT executed). No alert dialog.

TC-SEC-05 CSRF Token Validation

Field	Value
Priority	P1
Steps	1. Submit form with missing/invalid CSRF token
Expected	Request rejected: 403 Forbidden. CSRF token is required for state-changing requests.
Tool	REST Assured / OWASP ZAP

TC-SEC-06 Password Complexity Enforcement

Field	Value
Priority	P1
Test Data	password = "short" (fails all rules)
Steps	1. Navigate to password change/setup 2. Enter weak password
Expected	Error: "Password must be at least 8 characters, contain 1 uppercase, 1 lowercase, 1 number, 1 special character."

Non-Functional “ Cross-Browser & Responsive “

Test Cases

File: nfr/TC-NFR-CrossBrowser.md | Module: Cross-Browser | Total: 6 TC | Automated: 0

TC-CBR-01 “ Login Flow Across All Browsers

Field	Value
Priority	P0
Browsers	Chrome, Firefox, Edge, Safari
Steps	1. Execute full login flow (TC-AUTH-001 through TC-AUTH-005) on each browser
Expected	Consistent behavior: same error messages, same UI layout, same page flow. No browser-specific rendering bugs.

TC-CBR-02 “ A/B Test Creation Across Browsers

Field	Value
Priority	P0
Browsers	Chrome, Firefox, Edge, Safari
Steps	1. Execute TC-ABT-001 through TC-ABT-003 on each browser
Expected	Visual Editor loads correctly. Variation preview renders. Drag-and-drop works.

TC-CBR-03 “ Heatmap Rendering Across Browsers

Field	Value
Priority	P1
Browsers	Chrome, Firefox, Edge, Safari
Expected	Heatmap overlay renders identically. Color gradients match. Tooltips work.

TC-CBR-04 “ Dashboard Layout at 1920Ã—1080 (Desktop)

Field	Value
Priority	P0
Steps	1. Resize browser to 1920Ã—1080 â’ 2. Login â’ 3. Observe dashboard
Expected	Full layout: sidebar visible, experiment cards in grid, summary cards in row, no horizontal scroll.

TC-CBR-05 “ Dashboard Layout at 768Ã—1024 (Tablet)

Field	Value
Priority	P1
Browsers	Chrome, Safari
Steps	1. Resize to 768—1024 â€² 2. Observe responsive breakpoints
Expected	Sidebar collapses to hamburger. Cards stack vertically. Touch targets â‰¥ 44px. No content cut off.

TC-CBR-06 â€¢ Dashboard Layout at 375—812 (Mobile)

Field	Value
Priority	P1
Browsers	Chrome, Safari (iPhone)
Expected	Single column layout. Experiment cards full width. CTAs still tappable. Font sizes readable (â‰¥ 16px).

Non-Functional â€” Accessibility (WCAG 2.1 AA) â€” Test Cases

File: nfr/TC-NFR-Accessibility.md | Module: Accessibility | Total: 5 TC | Automated: 0

TC-ACC-01 â€¢ All Images Have Alt Text

Field	Value
Priority	P1
Criteria	Every element has a non-empty alt attribute. Decorative images have alt="".
Tool	axe-core / Lighthouse

TC-ACC-02 â€¢ Color Contrast Ratio â‰¥ 4.5:1

Field	Value
Priority	P1
Criteria	All text (including placeholder text) has contrast ratio â‰¥ 4.5:1 against background. Large text (â‰¥ 18px bold or â‰¥ 24px) has â‰¥ 3:1.
Tool	axe-core / Colour Contrast Analyser

TC-ACC-03 â€¢ Keyboard Navigation

Field	Value
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Priority	P0
Criteria	All interactive elements reachable via Tab key. Visible focus indicator (outline/ring). No keyboard traps. Logical tab order (leftâ†’right, topâ†’bottom).
Steps	1. Tab through entire login flow â†’ 2. Tab through experiment creation flow â†’ 3. Tab through reports
Expected	Every button, link, input, dropdown, and toggle is reachable and actionable via keyboard alone.

TC-ACC-04 â€¢ Form Labels Associated with Inputs

Field	Value
Priority	P1
Criteria	Every <input>, <select>, <textarea> has an associated <label> (via for attribute or wrapping). No placeholder-only labeling.
Tool	axe-core

TC-ACC-05 â€¢ ARIA Landmarks Present

Field	Value
Priority	P1
Criteria	Page has: <header role="banner">, <nav role="navigation">, <main role="main">, <footer role="contentinfo">. Dynamic regions have aria-live attributes.
Tool	axe-core